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Production of desalinated water and electricity in a dual-purpose plant operating in a dispatchable electricity system — techno-economical analysis

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Abstract

A scarcity of potable water in Israel is a major concern for the Israeli government. In the next 10 years, extensive planning will be done to add ~400–600 million m³ of potable water annually, mainly by building large seawater desalination plants. Furthermore, Israel's demand for electricity has escalated during the last decade at an average annual rate of 7.2%. This extremely high growth rate represents adding 500 MW_e of electrical capacity each year in the coming years. As a result, the Israeli market represents a unique situation where large electrical capacity and large water quantities should be added simultaneously. Most of the added electricity capacity will be accomplished through new modern gas turbine combined cycles (GTCC). These GTCC will be operated as intermediate load units. The present work analyzes the techno-economic characteristics of two types of dual-purpose plants that differ by seawater desalination technology: seawater reverse osmosis and low-temperature multi-effect distillation. The power generation in the assessed alternatives is modern GTCC. Two scenarios were studied: Scenario A, where the dual-purpose plant operates according to daily optimal dispatch policy from the electricity production point of view and the desalination plant operates at its nominal capacity for the period; and Scenario B, where the dual-purpose plant operates as a base-load unit (electricity and water). The water production cost was calculated for the two scenarios and for the two technological variants of the dual-purpose plant, while taking into account the economic influence of non-optimal power generation represented by Scenario "B".

Keywords: Seawater desalination; System optimization; Dual-purpose plant; Techno-economics

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1. Introduction

Seawater desalination plants require considerable energy resources. Consequently, they are frequently associated with power plants, as in the case of the Arab Gulf countries where the majority of desalinated water is produced in dual-purpose plants.

The profitability of dual-purpose plants from the national economy's point of view is usually dependent on its impact on power system expenditures throughout the lifetime of the plant. For this reason a socially optimal choice of the proper desalination alternative should be evaluated in a general framework of commonly accepted long-term electricity generation expansion analysis.

The present work analyses the production costs of two types of dual-purpose plants that produce electricity and water. The first one is a gas-fired combined cycle gas turbine (CCGT) integrated with low-pressure multi-effect distillation–thermal vapor compression (MED–TVC) sea water desalination plant; the second analyzed option is a gas-fired CCGT integrated with a seawater reverse osmosis (SWRO) plant. For the two options the electricity and the water price is presented.

It will be shown that the profitability of the above dual-purpose plants depends strongly on two main factors: the fuel mixture used to generate electricity for a specific market system and the power factor.

2. Description of analyzed alternatives

2.1. Alternative A

The dual-purpose plant contains a single-shaft gas-fired CCGT (technology F). The steam exits the low-pressure turbine and is directed to the MED modules at a pressure of 0.35 bar.

In order to keep the water production rate constant at low unit loads, high-grade steam (20 bar) is extracted from the steam turbine

(controlled extraction) and fed to the desalination plant. In these cases, part of the desalination modules are operated in MED mode with low-pressure steam and part of the modules are fed with high-pressure steam in the TVC mode. This configuration allows keeping the water production rate constant at all power station loads. Technical data for Alternative A are presented in Table 1 in the Appendix.

Alternative B

This dual-purpose plant contains single-shaft CCGT (technology 'F') and a SWRO plant. The electrical power required to operate the desalination plant is supplied directly from the power plant. During the periods when the power plant is not in operation, the SWRO consumes electricity from the national grid at a price similar to other industrial consumers. Technical data for alternative B are presented in Table 1 in the Appendix.

3. Methodology of analysis

The analysis aims to identify the lowest costs associated with electricity and water production from the point of view of the national economy. Note that here, only the electricity and water generation expenses were accounted for, while the costs related to transmission and distribution of electricity and water were not.

The different dual-purpose plants affecting operating parameters of a specific generating unit (in this case CCGT), are mainly net capacity, heat rate and availability. These changes could have a significant impact on the priority order according to which this unit is incorporated into unit commitment and an optimal dispatch process of the entire power system. Indeed, a loss of capacity, increasing heat rate or self-consumption, or variations in operation and maintenance costs may reduce the optimal number of operating hours of the considered generating unit, making it less profitable and noncompetitive

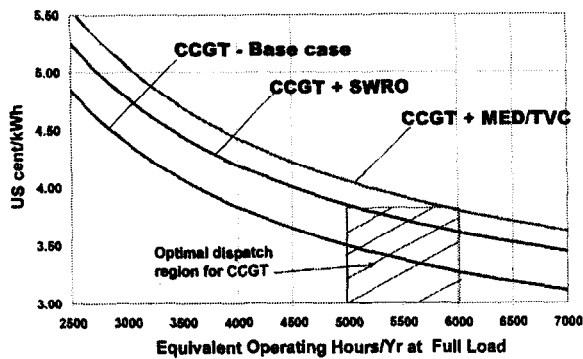


Fig. 1. Expected electricity production prices, US cent/kWh_{net}.

compared to other planning alternatives. This is illustrated in Fig. 1 where the different electricity generation costs are presented.

In our work, the expenses of the electricity system were calculated for the two alternatives presented above. These costs were compared to the cost of electricity production when no desalination plants are incorporated with the electricity generation plants: the Base Case. The added expenses (accumulated and discounted) associated with the dual-purpose plant as compared to the Base Case is allocated to the desalination process. This “penalty” is added to the expenses of the seawater desalination plant (capital recovery, operating and maintenance costs). This pricing concept is illustrated in Fig. 2.

As presented in Fig. 1, the optimal operating regime for the combined cycle plants (“pure” electricity plants as well as dual-purpose plants) lie in the range of 6500 h annually at full load. These will, of course, limit the profitability of dual-purpose plants containing distillation processes. On the other hand, compulsory, continuous operation of the unit in accordance with the needs of the desalination process, approximately 7000 h annually at full load, will increase the “penalty” to the power system even more. Therefore, the question is if these “penalties” to the electricity system exceed the benefits gained by the desalination system?

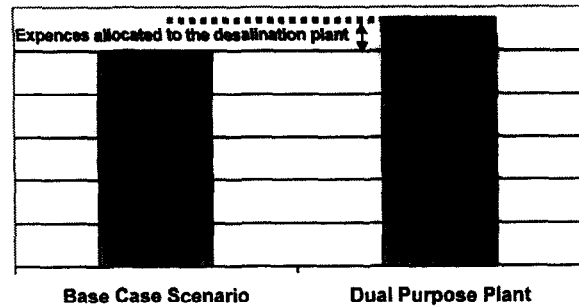


Fig. 2. Expenses allocated to the desalination system due to integration of the dual-purpose plant into the electricity generation system.

As will be demonstrated in the Results section, the answer is not conclusive. It strongly depends on fuel prices (difference between coal and natural gas), structure of the electricity generation system, electricity demand profile along the year, and type of available generating units. In our work, the technical characteristics of the dual-purpose plant were simulated using the commercial Thermoflow[®] software package. The electricity system analysis was conducted using the UCOD simulation package developed by the Israel Electricity Corporation. The approach used in UCOD is based on the simulation of random outage conditions, thus providing a proper response to meet the requirements of a specific generating system. UCOD allows an expansion plan to be evaluated over a multi-year planning horizon. The results provide an overview of the system production costs and the operation of each generating resource.

4. Results

In this section we present electricity and water production prices associated with the two types of dual-purpose plants. It is important to emphasize again that in all the analyzed cases the amount of annual electricity produced and water produced is identical.

A dual-purpose plant is less efficient than the original CCGT from the electricity system point of view (higher net heat rate); as a result, its priority for operation will be lower as compared to the single-purpose CCGT (lower marginal cost). On the other hand, the considerable investment required for the desalination plant can be justified only if the plant is operating continuously during long periods of time. This requires almost a base-load operation of the associated CCGT, which will force the electricity system dispatcher not to operate the dual-purpose plants at its optimal priority from the electricity system point of view.

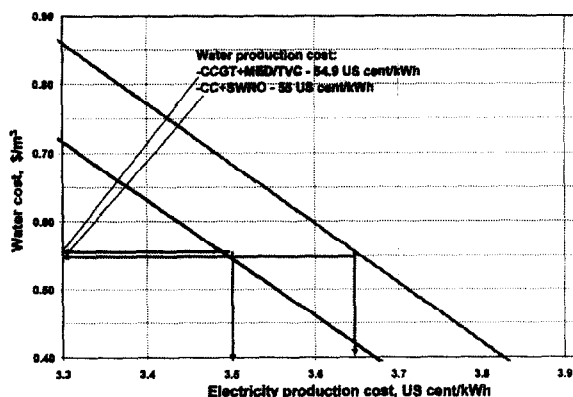


Fig. 3. Electricity–water allocation, Scenario A.

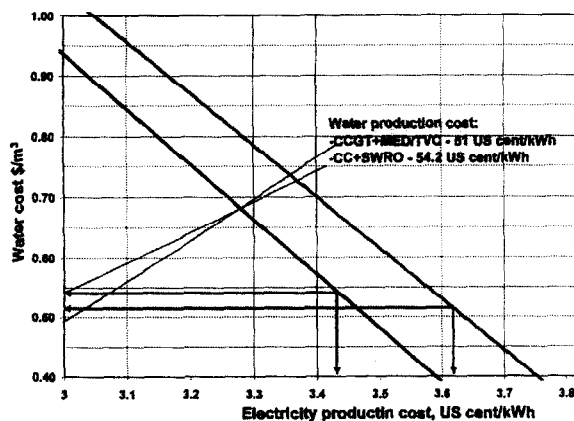


Fig. 4. Electricity–water allocation, Scenario B.

Base-load operation of the dual-purpose plants will allocate part of the electricity produced in the low-cost, base-load, coal-fired units to the dual-purpose plants, which have higher marginal costs compared to the coal-fired plant and to other CCGT plants in the system. This will result in a penalty to the national power system. As discussed above, this penalty is allocated to the water production cost. Consequently, the costs associated with the electricity system are similar for all the alternatives. This approach simplifies the analysis and allows us to treat the “water production side” of the dual-purpose plant separately.

We analyzed two different scenarios, distinguished by the utilization level of the CCGT.

- Scenario A (Fig. 3) — the reference CCGT plant as well as Alternatives A and B are optimally dispatched at 6550 h/year at full load.
- Scenario B (Fig. 4) — the dual-purpose plants as well as the reference plants operate at an elevated utilization factor, yielding 7000 h/year at full load.

5. Discussion and conclusions

1. Presently, a lack of indigenous energy resources caused Israel to base its energy production on coal. Currently, coal plants, which constitute about a half of the total installed capacity, produce more than 70% of electricity in Israel (see Fig. 5). A commercial amount of natural gas found in Israeli territorial waters in 2000, as well as large reserves located in close vicinity of Israel, considerably affected its electricity generation expansion plans. Natural gas-fired CCGT became the main planning alternative to meet a rapidly growing demand for electricity.

2. As illustrated in Fig. 5, the existing coal-fired plants operate as pure base-load units, resulting in less “room” remaining for base-load

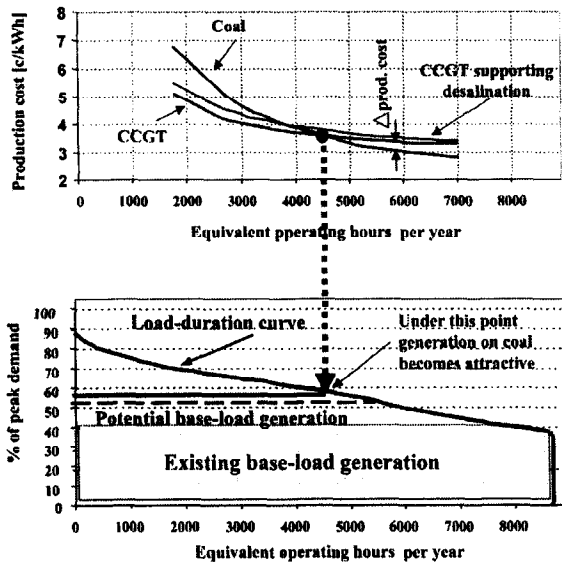


Fig. 5. Typical load duration curve for the Israeli electricity system.

operation of the CCGT+MED/TVC (Alternative A). Thus, forcing Alternative A to operate as a base-load unit will reduce the amount of energy which can be produced in the coal-fired plants, which have considerably lower marginal costs.

3. The projected prices within this study are 51–55 US cent/m³, which are comparable to the latest prices in the sea water desalination market in Israel (52 US cent/m³ for a 100 m³/year plant in Ashkelon). It may be safely assumed that these prices may be even lower when considering the impressive development in seawater desalination technologies.

4. The water production costs assuming Scenario A are similar for both alternatives, with little advantage to Alternative A (CCGT+MED–TVC). This is due mainly to the allocated penalty associated with the difference in fuel prices (gas vs. coal). This scenario represents the present situation in the Israeli electricity market where optimal dispatch of the gas-fired CCGT is about 6500 h/year at full load.

5. In time, the share of inexpensive base-load electricity production by coal is expected to decrease in favor of gas-fired CCGT (represented by Scenario B). Here, the penalty associated with non-optimal dispatch of the electricity system is greatly reduced, resulting in reduction of the water price for Alternative A.

Appendix

Table 1
Dual-purpose plant technical data

Data for the electricity generating unit	CCGT reference plant	CCGT + MED/TVC	CCGT+ SWRO
Installed capacity, MW _e	369	348	369
Investment, US \$/kW _{installed}	540	605	540
Self-consumption including desalination plant, %	2.30	3.69	6.37
Gross heat rate, kcal/kWh	1,532	1,644	1,532
Gross efficiency, %	56.13	52.31	56.13
Fuel low heating value, kcal/kg	11,500	11,500	11,500
Gross fuel specific consumption, kg/kWh	0.133	0.143	0.133
Average fuel cost, \$/MBtu	3.55	3.55	3.55
Fixed O&M, \$/kW/y	17	18.0	17
Variable O&M, US ¢/kWh	0.120	0.13	0.12
Annual O&M escalation, %	0.0	0.0	0.0
Equivalent forced outage rate, %	7.00	7.0	7.0
Scheduled maintenance, d/y	29	29	29
Equivalent operating hours at full load, h/y	6,550	7,000	6,550

Seawater desalination plant		
Technology	SWRO	MED-TVC
Investment, US \$/m ³ /d	800	850
Basic module size, m ³ /d	8,000	18,000
Number of modules	10	5
Installed capacity, m ³ /d	80,000	90,000
Steam flow from low-pressure turbine (nominal conditions), MED mode, t/h	NA	364
Steam pressure for MED mode, bar (saturated)	NA	0.35
Economic ratio MED mode, m ³ water/t steam	NA	10
Economic ratio TVC mode, m ³ water/t steam	NA	14
Steam flow for NCG removal, t/h	NA	7.5
Steam flow TVC mode/module, t/h	NA	52
Nominal water production rate, m ³ /h	3,333	3,640
Annual production rate, 10 ⁶ m ³ /y	27.5	27.5
Specific electricity consumption, kWh/m ³	4	1.2
Self-electricity consumption, MW	13.3	4.37
Chemical cost, US ¢/m ³	5	5
Membrane replacement cost, US ¢/m ³	6.3	NA